

Year 7 · Chapter 10 · Analysing data: answers (teacher copy)

10.01 Interpreting graphs (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: How many chose Tennis? → 10
- Column 1, E2: How many chose Dance? → 3
- Column 1, E3: between 25 and 30 → 27.5
- Column 2, E1: From January to April → January: 30°C, April: 23°C; change: 7°C down
- Column 2, E2: From February to September → February: 29°C, September: 21°C; change: 8°C down
- Column 3, E1: How many more chose Tennis than Dance? What percentage of the class is that (1 d.p.)? → $10 - 3 = 7$; $7 \div 46 \times 100 = 15.2\%$
- Column 3, E2: What fraction, and what percentage (1 d.p.), chose Tennis? → 10 out of 46; fraction: $\frac{5}{23}$; percentage: $10 \div 46 \times 100 = 21.7\%$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-----------|-------------------|-------------------|
| 1. 17 | 10. 85 | 18. 19°C to 30°C: |
| 2. 3 | 11. 110 | up 11°C |
| 3. 13 | 12. 22 | 19. 18°C to 19°C: |
| 4. 46 | 13. 70 | up 1°C |
| 5. Soccer | 14. 75 | 20. 16°C to 24°C: |
| 6. 3 | 15. 7 | up 8°C |
| 7. 170 | 16. 26°C to 29°C: | 21. 18°C to 26°C: |
| 8. 11 | up 3°C | up 8°C |
| 9. 45 | 17. 23°C to 16°C: | 22. 21°C to 18°C: |
| | down 7°C | down 3°C |

10.02 Misleading graphs (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: The vertical axis starts at 0. → F
- Column 1, E2: The vertical axis starts at 90. → M
- Column 2, E1: The y-axis goes 0, 5, 10, 50, 100. → problem: uneven scale; fix: use equal steps
- Column 2, E2: The y-axis starts at 1000. → problem: axis does not start at 0; fix: start the axis at 0 (or show a break)
- Column 3, E1: A = 92, B = 96, axis starts at 90 → drawn: $6 \div 2 = 3$; real: $96 \div 92 = 1$
- Column 3, E2: A = 51, B = 54, axis starts at 50 → drawn: $4 \div 1 = 4$; real: $54 \div 51 = 1.1$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------|---------------------|--------------------|
| 1. M | 13. picture looks 4 | 17. looks 5 times; |
| 2. F | times as big; | really 1.4 times |
| 3. M | enlarge in one | 18. looks 2 times; |
| 4. M | direction only | really 1 times |
| 5. F | 14. missing labels; | 19. looks 5 times; |
| 6. M | label both axes | really 1.1 times |
| 7. F | 15. data left out; | 20. looks 4 times; |
| 8. M | show every | really 1.3 times |
| 9. F | month | 21. looks 4 times; |
| 10. M | 16. front sectors | really 1.1 times |
| 11. F | look bigger; | |
| 12. M | draw it flat | |

10.03 Dot plots (booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: 7, 8, 6, 4, 8, 8, 3 → 8
- Column 1, E2: 8, 4, 1, 6, 4, 7, 4 → 4
- Column 2, E1: How many values? → count from the plot; = 17
- Column 2, E2: What is the mode? → count from the plot; = 6

- Column 3, E1: 5, 1, 4, 8, 5, 4, 4, 6, 8, 3 → ordered: 1, 3, 4, 4, 4, 5, 5, 6, 8, 8; mode: 4; range: $8 - 1 = 7$
- Column 3, E2: 1, 3, 8, 4, 6, 5, 8, 5, 5, 1 → ordered: 1, 1, 3, 4, 5, 5, 5, 6, 8, 8; mode: 5; range: $8 - 1 = 7$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------|--------|-------------------|
| 1. 2 | 17. 3 | 33. 14 |
| 2. 8 | 18. 3 | 34. 7 |
| 3. 2 | 19. 5 | 35. mode 3 and 5, |
| 4. 9 | 20. 1 | range 6 |
| 5. 7 | 21. 4 | 36. mode 6, range |
| 6. 8 | 22. 7 | 6 |
| 7. 2 | 23. 9 | 37. mode 1 and 3 |
| 8. 3 | 24. 5 | and 6, range 6 |
| 9. 2 | 25. 8 | 38. mode 3, range |
| 10. 1 | 26. 6 | 6 |
| 11. 6 | 27. 2 | 39. mode 1, range |
| 12. 3 | 28. 3 | 7 |
| 13. 4 | 29. 9 | 40. mode 1, range |
| 14. 4 | 30. 4 | 7 |
| 15. 1 | 31. 11 | 41. mode 2 and 6, |
| 16. 2 | 32. 3 | range 6 |

10.04 Stem-and-leaf plots (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: 3 | 9 → 39
- Column 1, E2: 2 | 3 → 23
- Column 2, E1: How many scores are there? → count the leaves; = 15
- Column 2, E2: What is the lowest score? → first leaf, first row; = 22
- Column 3, E1: 41, 35, 25, 20, 41, 26, 45, 43 → ordered: 20, 25, 26, 35, 41, 41, 43, 45; 2 | 0 5 6; 3 | 5; 4 | 1 1 3 5
- Column 3, E2: 31, 41, 12, 42, 23, 27, 33, 21 → ordered: 12, 21, 23, 27, 31, 33, 41, 42; 1 | 2; 2 | 1 3 7; 3 | 1 3; 4 | 1 2

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------|--------|--------------------------|
| 1. 18 | 16. 37 | 31. 62 |
| 2. 44 | 17. 53 | 32. 40 |
| 3. 50 | 18. 73 | 33. 1 |
| 4. 25 | 19. 86 | 34. 5 |
| 5. 74 | 20. 69 | 35. 1 5 8; 2 0 1 2; |
| 6. 38 | 21. 12 | 3 4; 4 4 6 |
| 7. 43 | 22. 94 | 36. 1 7 9; 2 2 4; 3 |
| 8. 20 | 23. 76 | 5 5; 4 2 8 |
| 9. 85 | 24. 61 | 37. 2 3; 3 0 2 2 3 4 |
| 10. 64 | 25. 46 | 6 6 |
| 11. 99 | 26. 55 | 38. 2 3 4 8 9; 3 0 |
| 12. 91 | 27. 47 | 2 9; 4 1 |
| 13. 56 | 28. 92 | 39. 1 7 9; 2 2 8; 3 |
| 14. 34 | 29. 72 | 5; 4 1 6 8 |
| 15. 15 | 30. 93 | |

10.05 Mean and mode (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: 6, 9, 14, 11 → 10
- Column 1, E2: 12, 5, 20, 7, 11 → 11
- Column 2, E1: 25, 39, 39, 22 → total: $25 + 39 + 39 + 22 = 125$; mean: $125 \div 4 = 31.3$
- Column 2, E2: 19, 20, 40, 30, 24, 29 → total: $19 + 20 + 40 + 30 + 24 + 29 = 162$; mean: $162 \div 6 = 27$
- Column 3, E1: 5, 1, 1, ? (mean 7) → total must be $7 \times 4 = 28$; known: 7; missing: $28 - 7 = 21$

6. Column 3, E2: 13, 22, 1, 31, ? (mean 17) → total must be $17 \times 5 = 85$; known: 67; missing: $85 - 67 = 18$

Questions (in order: Easy, Medium, Challenging)

1. 9	17. 16	33. 16.8
2. 11	18. 13	34. 23.6
3. 12	19. 12	35. 17
4. 12	20. 12	36. 28.4
5. 9	21. 13	37. 20.4
6. 11	22. 12	38. 18
7. 12	23. 12	39. 23.7
8. 13	24. 10	40. 15
9. 11	25. 8	41. 16
10. 17	26. 13	42. 11
11. 7	27. 8	43. 11
12. 8	28. 10	44. 16
13. 6	29. 11	45. 16
14. 12	30. 26.3	46. 7
15. 9	31. 18	
16. 11	32. 23	

10.06 Median and range (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: 16, 21, 4, 18, 4 → 16
- Column 1, E2: 13, 11, 22, 14, 13 → 13
- Column 1, E3: 8, 38, 38, 28, 14 → 30
- Column 2, E1: 24, 27, 13, 36, 31, 26 → ordered: 13, 24, 26, 27, 31, 36; $(26 + 27) \div 2 = 26.5$
- Column 2, E2: 15, 18, 14, 8, 21, 14 → ordered: 8, 14, 14, 15, 18, 21; $(14 + 15) \div 2 = 14.5$
- Column 3, E1: 28, 22, 39, 24, 31, 25, 26, 36 → ordered: 22, 24, 25, 26, 28, 31, 36, 39; median: 27; range: $39 - 22 = 17$
- Column 3, E2: 60, 57, 30, 54, 39, 24, 24 → ordered: 24, 24, 30, 39, 54, 57, 60; median: 39; range: $60 - 24 = 36$

Questions (in order: Easy, Medium, Challenging)

1. 11	17. 27	33. 9
2. 16	18. 42	34. 28
3. 7	19. 40	35. 36
4. 18	20. 24	36. median 45, range 40
5. 13	21. 36	37. median 31, range 27
6. 9	22. 37	38. median 40, range 48
7. 12	23. 38	39. median 39.5, range 24
8. 18	24. 25	40. median 42, range 49
9. 10	25. 36	41. median 26, range 47
10. 14	26. 22	42. median 29, range 19
11. 13	27. 14.5	
12. 10	28. 20.5	
13. 36	29. 11	
14. 16	30. 18	
15. 45	31. 24	
16. 25	32. 29	

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: How many values? → 11
- Column 1, E2: Lowest value → 36
- Column 1, E3: 3 | 0 5
4 | 8

5 | 7

6 | 2 7 → 37

4. Column 2, E1: Find the median. → 14 values: the middle ones; = 6.5

5. Column 2, E2: Find the mean (1 d.p.). → $91 \div 14$; = 6.5

6. Column 2, E3: 1 | 9

2 | 6 6 6

3 | 8

4 | 5

5 | 5 → 7 values: the 4th; median = 26

7. Column 3, E1: 14, 11, 10, 11, 15, 19, 44 → with: mean 17.7, median 14; without 44: mean 13.3, median 12.5; the mean changes most

8. Column 3, E2: 11, 11, 15, 13, 12, 13, 53 → with: mean 18.3, median 13; without 53: mean 12.5, median 12.5; the mean changes most

Questions (in order: Easy, Medium, Challenging)

1. 73	11. 24	15. mean 21.7 →
2. 37	12. 39	16.2; median 17
3. 50	13. mean 17.7 →	→ 16.5; the mean
4. 5	12.3; median 13	
5. 52	→ 12; the mean	16. mean 20.9 →
6. 21	14. mean 18.9 →	15.7; median 17
7. 50	14.8; median 16	→ 16; the mean
8. 27	→ 15; the mean	17. mean 18.9 →
9. 36		14.2; median 15
10. 8		→ 14.5; the mean

10.08 Comparing data (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: A: range 18, B: range 25 → A
- Column 1, E2: A: range 20, B: range 28 → A
- Column 2, E1: A: 13, 10, 10, 14
B: 4, 7, 2, 18 → A: $47 \div 4 = 11.75$; B: $31 \div 4 = 7.75$; higher mean: A
- Column 2, E2: A: 4, 16, 16, 18
B: 8, 7, 9, 13 → A: $54 \div 4 = 13.5$; B: $37 \div 4 = 9.25$; higher mean: A
- Column 3, E1: Class A: 7, 13, 15, 9, 7
Class B: 20, 14, 11, 17, 11 → A: mean 10.2, range 8; B: mean 14.6, range 9; B did better; A is more consistent
- Column 3, E2: Class A: 12, 19, 20, 13, 9
Class B: 20, 8, 7, 6, 15 → A: mean 14.6, range 11; B: mean 11.2, range 14; A did better; A is more consistent

Questions (in order: Easy, Medium, Challenging)

1. A	16. B	26. A mean 9.8, range 7; B mean 11.6, range 10
2. B	17. A	
3. A	18. B	
4. B	19. A	
5. B	20. A	27. A mean 13.4, range 11; B mean 11.2, range 12
6. B	21. A	
7. B	22. B	
8. A	23. A	28. A mean 10.8, range 15; B mean 7.6, range 6
9. A	24. B	
10. B	25. A mean 10.4, range 7; B mean 11.4, range 15	29. A mean 12.8, range 11; B mean 11.2, range 8
11. B		
12. A		
13. B		
14. A		
15. B		